THE COVERING RADIUS OF THE FIRST-ORDER REED–MULLER CODE IN NINE VARIABLES IS 242

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Abstract. We prove that the covering radius of the binary first-order Reed–Muller code $\mathrm{RM}(1,9)$ is exactly 242. Equivalently, the maximum nonlinearity of a Boolean function in nine variables is 242. Our proof combines a one-point rigidity theorem with a spectral and computational exclusion of the two larger candidate nonlinearities. We first show that every nine-variable Boolean function of nonlinearity 243 admits a unique one-point modification that raises its nonlinearity to 244: the point to be changed is the vector sum of its support. This part uses exact Walsh-update identities, integer interpolation, rigidity of a quadratic residue, and a sharp Walsh ℓ^1 estimate for cubic Boolean functions. To exclude nonlinearity 244, we use Parseval's identity and congruences of integral Walsh coefficients to force an exceptionally restricted spectrum. An associated cubic residue phase must attain the extremal Walsh ℓ^1 bound 10240. We eliminate one equality spectrum by hyperplane slicing; the other forces the original Boolean function from the signs of the cubic phase and yields a double-complement-pairing identity for its cubic top part. The classification of alternating trilinear forms over $\mathbb{F}_2$, an exact filter of their representatives, and two exhaustive enumerations of 2^{18} quadratic supplements then rule out every remaining candidate. Together with the known construction attaining nonlinearity 242, these arguments establish the asserted covering radius.

1. Introduction

A *Boolean function* in n variables is a map $f : \mathbb{F}_2^n \to \mathbb{F}_2$. We identify f with its truth table, a binary word of length 2^n. For two Boolean functions f and g, their Hamming distance $d_H(f,g)$ is the number of inputs on which their values differ. The *nonlinearity* of f is the minimum distance from f to an affine Boolean function:

$$\mathrm{NL}(f) = \min_{\substack{a\in\mathbb{F}_2^n \\ b\in\mathbb{F}_2}} d_H\big(f, a\cdot x + b\big) = \min_{\substack{a\in\mathbb{F}_2^n \\ b\in\mathbb{F}_2}} \mathrm{wt}\big(f(x) + a\cdot x + b\big). \tag{1.1}$$

Here wt denotes Hamming weight, and addition inside the weight is in $\mathbb{F}_2$. Thus nonlinearity measures how far a Boolean function lies from every affine

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approximation. Large nonlinearity is a central criterion in the design of cryptographic Boolean functions because it limits their correlation with affine functions; it is equally natural from the standpoint of coding theory [2].

For a binary code $\mathcal{C} \subseteq \mathbb{F}_2^N$, its *covering radius* is the largest distance of a binary word from its nearest codeword:

$$\rho(\mathcal{C}) = \max_{u \in \mathbb{F}_2^N} \min_{c \in \mathcal{C}} d_H(u, c). \tag{1.2}$$

Equivalently, $\rho(\mathcal{C})$ is the smallest integer R such that the Hamming balls of radius R centred at the codewords cover the entire ambient space $\mathbb{F}_2^N$. The binary first-order Reed–Muller code

$$\mathrm{RM}(1, n) = \left\{ \left(a \cdot x + b\right)_{x \in \mathbb{F}_2^n} : a \in \mathbb{F}_2^n,\ b \in \mathbb{F}_2 \right\} \subseteq \mathbb{F}_2^{2^n}$$

consists precisely of the truth tables of affine Boolean functions. Every word in $\mathbb{F}_2^{2^n}$ is the truth table of exactly one Boolean function on $\mathbb{F}_2^n$. Consequently, substituting this description of $\mathrm{RM}(1, n)$ into the covering-radius definition and comparing with (1.1) gives the fundamental identity

$$\boxed{\rho\big(\mathrm{RM}(1, n)\big) = \max_{f : \mathbb{F}_2^n \to \mathbb{F}_2} \mathrm{NL}(f).} \tag{1.3}$$

In other words, determining the covering radius of $\mathrm{RM}(1, n)$ and determining the maximum possible nonlinearity of an n-variable Boolean function are *exactly the same extremal problem*.

The difficulty of this problem is particularly pronounced for odd n. The Walsh transform

$$W_f(a) = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x) + a \cdot x}$$

provides the equivalent spectral expression

$$\mathrm{NL}(f) = 2^{n-1} - \frac{1}{2} \max_{a \in \mathbb{F}_2^n} |W_f(a)|. \tag{1.4}$$

When n is even, bent functions achieve a perfectly flat spectrum, providing an extremal benchmark. No such spectrum is possible when n is odd, since Parseval's identity would require all Walsh magnitudes to be the nonintegral number $2^{n/2}$. The resulting optimization is global: a large nonlinearity requires the entire Walsh spectrum, rather than one or a few coefficients, to remain small.

The case $n = 9$ illustrates how hard this global constraint can be. Appending one variable to an eight-variable bent function yields nonlinearity 240; constructions of Kavut, Maitra and Yücel [5] attained 241, and Kavut and Yücel [6] obtained functions with nonlinearity 242. On the other hand, Walsh integrality and Parseval's identity imply the upper bound 244. The long-standing uncertainty was therefore confined to the exceptionally narrow interval

$$242 \leq \rho\big(\mathrm{RM}(1, 9)\big) \leq 244. \tag{1.5}$$

Closing this gap is not a small-scale search problem. There are 2^{512} nine-variable truth tables; changing one output bit affects all 512 Walsh coefficients at once.

Our principal result settles the problem.

Theorem 1.1 (Main theorem)**.** *The covering radius of the binary first-order Reed–Muller code* $\mathrm{RM}(1,9)$ *is*

$$\boxed{\rho(\mathrm{RM}(1,9)) = 242.}$$

Equivalently, every nine-variable Boolean function has nonlinearity at most 242*, and this value is attained.*

A key feature of our approach is a precise connection between the two potential improvements beyond 242. For $t \in \mathbb{F}_2^9$, let δ_t be the indicator of the singleton $\{t\}$, and define the *support sum* of a Boolean function by

$$p(f) = \sum_{x \in \mathrm{supp}(f)} x \in \mathbb{F}_2^9,$$

where the sum is taken in the vector space $\mathbb{F}_2^9 = \mathbb{F}_2^9$.

Theorem 1.2 (One-point rigidity)**.** *If* $\mathrm{NL}(f) = 243$*, then*

$$\mathrm{NL}(f + \delta_{p(f)}) = 244,$$

and $p(f)$ *is the unique point whose flip increases nonlinearity to* 244*. Conversely, if* $\mathrm{NL}(g) = 244$*, then* $\mathrm{NL}(g + \delta_t) = 243$ *for every* $t \in \mathbb{F}_2^9$ *and* $p(g) = 0$*. Consequently the map*

$$\{g : \mathrm{NL}(g) = 244\} \times \mathbb{F}_2^9 \longrightarrow \{f : \mathrm{NL}(f) = 243\}, \qquad (g, t) \longmapsto g + \delta_t$$

is a bijection.

We prove the one-point theorem by exploiting the integer structure of Walsh coefficients. Exact interpolation shows that successive binary residues of a normalized Walsh spectrum are Boolean polynomials of small degree. A quadratic residue rigidity argument and a sharp ℓ^1 estimate for cubic spectra then identify the unique improving flip. It follows that nonlinearity 243 cannot occur unless nonlinearity 244 does; hence it remains to exclude the latter.

Suppose, therefore, that a Boolean function of nonlinearity 244 exists. Parseval's identity and integral Walsh congruences force Walsh magnitudes 8 and 24, occurring 64 and 448 times, respectively. The next binary residue yields a cubic Boolean phase h for which $\|W_h\|_1 \geq 10240$. Our cubic spectral inequality shows that equality must hold and leaves only two possible absolute spectra. We eliminate one by comparing the Walsh energies on the two sides of an eight-dimensional hyperplane. In the remaining spectrum, the original function is determined by the signs of W_h; comparison of its degree-six coefficients with the cubic top part ω of h gives a nonlinear double-complement-pairing identity $\omega^{\#\#} = \omega$. The classification of nine-dimensional

alternating trilinear forms [1, 4] combined with an exact filter of the fixed-point identity, then leaves only two cubic representatives. For each of them, nine linear face constraints and nine phase-translation directions reduce the 36 quadratic coefficients to a complete 18-dimensional transversal. Two exact 2^{18}-state enumerations exclude every remaining equality-spectrum candidate.

We obtained the conclusion $\rho(\mathrm{RM}(1,9)) = 242$ on September 30, 2026. The present paper gives our independent proof of this result. Contemporaneous results were also obtained by Wang, Lou and Wang [8] and by Hou, Sabo and Slaughter [3]. Our approach combines one-point rigidity, arithmetic restrictions on integral Walsh spectra, and sharp estimates for cubic Boolean functions. The classification of alternating trilinear forms reduces the remaining obstruction to two cubic representatives; linear constraints and translation symmetries then leave 2^{18} quadratic supplements for each. An exhaustive computer calculation eliminates these finite possibilities and completes the proof.

2. Walsh and integer-polynomial preliminaries

Set $V = \mathbb{F}_2^9$ and $[9] = \{1, \dots, 9\}$. For $S \subseteq [9]$ write $x^S = \prod_{i \in S} x_i$, with $x^\varnothing = 1$, and write $S^c = [9] \setminus S$. The algebraic normal form (ANF) of a Boolean function is its unique squarefree polynomial over $\mathbb{F}_2$. Integer sums below are explicitly distinguished from sums in $\mathbb{F}_2$.

For a Boolean function f, define its sign and unnormalized Walsh transform by

$$s_f(x) = (-1)^{f(x)}, \qquad W_f(a) = \sum_{x \in V} (-1)^{f(x)+a\cdot x}, \qquad a \in V.$$

Character orthogonality gives

$$\sum_{a \in V} (-1)^{a\cdot x} W_f(a) = 512 s_f(x), \tag{2.1}$$

$$\sum_{a \in V} W_f(a)^2 = 512^2 = 262144, \tag{2.2}$$

$$\mathrm{NL}(f) = 256 - \tfrac{1}{2}\|W_f\|_\infty. \tag{2.3}$$

In particular $\|W_f\|_\infty \geq 24$ because this maximum is an even integer and $\sqrt{512} > 22$. This proves the upper bound 244 in (1.5). Symmetry of the Walsh–Hadamard matrix also yields

$$\sum_{a \in V} s_h(a) W_f(a) = \sum_{x \in V} s_f(x) W_h(x). \tag{2.4}$$

For any $t \in V$, a direct change in one sign gives the exact update

$$W_{f+\delta_t}(a) = W_f(a) - 2 s_f(t) (-1)^{a\cdot t}. \tag{2.5}$$

It follows that $|\,\mathrm{NL}(f + \delta_t) - \mathrm{NL}(f)| \leq 1$.

2.1. **An integer Walsh polynomial.** Every integer-valued function on $\{0,1\}^9$ has a unique integer multilinear representation, by Möbius inversion:

$$T(a) = \sum_{S\subseteq[9]} T_S a^S, \qquad T_S = \sum_{U\subseteq S} (-1)^{|S|-|U|} T(\mathbf{1}_U). \tag{2.6}$$

In particular, if m divides every value of T, then it divides every coefficient T_S. This fact justifies division in $\mathbb{Z}$ *before* reduction modulo two.

For $S \subseteq [9]$, let

$$M_S(f) = \sum_{\substack{x\in V \\ x_i=1\ (i\in S)}} f(x)$$

be the integer weight on the indicated coordinate face. Expanding $(-1)^{a\cdot x} = \prod_{i:x_i=1}(1-2a_i)$ shows that the coefficient $C_S(f)$ of a^S in $W_f(a)$ is

$$C_S(f) = (-1)^{|S|}512 - 2(-2)^{|S|} M_S(f). \tag{2.7}$$

For $S = \varnothing$ this reads $W_f(0) = 512 - 2\,\mathrm{wt}(f)$. For $|S| \geq 3$ the right-hand side is divisible by 16, and for $|S| \geq 4$ it is divisible by 32.

Lemma 2.1 (Integral residue degrees)**.** *If $W_f(a) \equiv c \pmod 8$ at every frequency, then $((W_f - c)/8) \bmod 2$ has degree at most two. If $W_f(a) \equiv c \pmod{16}$, then $((W_f - c)/16) \bmod 2$ has degree at most three.*

Proof. Both quotients are integer-valued, so their coefficients are obtained by exact division of the integer coefficients in (2.6). Equation (2.7) makes all coefficients of degree at least three even after division by 8, and all coefficients of degree at least four even after division by 16. Reducing the already integral multilinear representation modulo two proves the assertions. □

We recall the standard Reed–Muller minimum-weight estimate and give a proof for completeness.

Lemma 2.2 (Reed–Muller minimum weight)**.** *Let $m \geq 1$ and $0 \leq d \leq m$. If $u : \mathbb{F}_2^m \to \mathbb{F}_2$ is nonzero and $\deg u \leq d$, then $\mathrm{wt}(u) \geq 2^{m-d}$.*

Proof. We argue by induction on the number $m \geq 1$ of variables. For $m = 1$, a nonzero Boolean function is either the constant function 1, of weight 2, or a nonconstant affine function, of weight 1. In both cases, the asserted inequality holds for all allowed values of d.

Now let $m \geq 2$ and assume the result for $m-1$ variables. If $d = 0$, then u is the nonzero constant function and $\mathrm{wt}(u) = 2^m = 2^{m-d}$. We may therefore take $d \geq 1$. Write $x = (x', z)$, with $x' \in \mathbb{F}_2^{m-1}$ and $z \in \mathbb{F}_2$. Separating the terms of the algebraic normal form according to whether they contain z gives

$$u(x', z) = u_0(x') + z\,u_1(x'), \qquad \deg u_0 \leq d, \quad \deg u_1 \leq d-1,$$

where addition is in $\mathbb{F}_2$. The restrictions to $z = 0$ and $z = 1$ are u_0 and $u_0 + u_1$, respectively. In particular,

$$\mathrm{wt}(u) = \mathrm{wt}(u_0) + \mathrm{wt}(u_0 + u_1).$$

If $u_1 = 0$, the two restrictions coincide. Since u is nonzero, so is u_0, and hence $\mathrm{wt}(u) = 2\,\mathrm{wt}(u_0)$. Apply the induction hypothesis to u_0 with degree bound $d' = \min\{d, m-1\}$. It follows that

$$\mathrm{wt}(u) \geq 2 \cdot 2^{(m-1)-d'} = 2^{m-d'} \geq 2^{m-d}.$$

Now suppose $u_1 \neq 0$. For every x' with $u_1(x') = 1$, the two values $u(x', 0) = u_0(x')$ and $u(x', 1) = u_0(x') + 1$ are different; therefore exactly one of them equals 1. Such an x' contributes one to $\mathrm{wt}(u)$, irrespective of $u_0(x')$. Consequently,

$$\mathrm{wt}(u) \geq \mathrm{wt}(u_1).$$

Since u_1 is a nonzero function in $m-1$ variables of degree at most $d-1$, the induction hypothesis yields

$$\mathrm{wt}(u) \geq \mathrm{wt}(u_1) \geq 2^{(m-1)-(d-1)} = 2^{m-d}.$$

Both cases establish the claim. □

Recall that if a Boolean function q has degree at most two, its quadratic polar form has even rank $r \leq 8$, and the nonzero entries of W_q have magnitude $2^{9-r/2}$ at precisely 2^r frequencies. Consequently

$$\|W_q\|_1 = 2^{9+r/2} \leq 8192. \tag{2.8}$$

This follows by squaring W_q and summing over the radical of the polar form; alternatively it is the standard quadratic Walsh formula [2].

3. The sharp Walsh norm bound for cubics

The following lemma supplies the common analytic input for both remaining nonlinearities.

Lemma 3.1 (Exact weight expansion)**.** *Let $u : \mathbb{F}_2^m \to \mathbb{F}_2$ be the XOR of its distinct ANF monomials with supports in a set $\mathscr{M}$, including the empty support when appropriate. Then, as an integer identity,*

$$\mathrm{wt}(u) = \sum_{\varnothing \neq \mathscr{A} \subseteq \mathscr{M}} (-2)^{|\mathscr{A}|-1} 2^{m-|\bigcup_{J \in \mathscr{A}} J|}. \tag{3.1}$$

If $m \geq 7$ and $\deg u \leq 3$, then $4 \mid \mathrm{wt}(u)$ and $8 \mid W_u(a)$ at every frequency. If $m \geq 4$ and $\deg u \leq 3$, then $\mathrm{wt}(u)$ is even.

Proof. Write the algebraic normal form of u as

$$u(x) = \sum_{J \in \mathscr{M}} x^J \pmod 2, \qquad x^J = \prod_{j \in J} x_j.$$

For a Boolean input x, each monomial x^J is either 0 or 1. Thus $(-1)^{x^J} = 1 - 2x^J$, and taking the sign of the XOR of the monomials gives the following identity of *ordinary integers*:

$$(-1)^{u(x)} = \prod_{J \in \mathscr{M}} (-1)^{x^J} = \prod_{J \in \mathscr{M}} (1 - 2x^J).$$

Expanding the right-hand product amounts to choosing, for each $J \in \mathscr{M}$, either 1 or $-2x^J$. Consequently,

$$(-1)^{u(x)} = 1 + \sum_{\varnothing \neq \mathscr{A} \subseteq \mathscr{M}} (-2)^{|\mathscr{A}|} \prod_{J \in \mathscr{A}} x^J.$$

On $\{0,1\}^m$ we have $x_i^2 = x_i$, so $\prod_{J\in\mathscr{A}} x^J = x^{\bigcup_{J\in\mathscr{A}} J}$. Moreover, the Boolean value $u(x)$, viewed as the integer 0 or 1, satisfies $u(x) = (1 - (-1)^{u(x)})/2$. Substituting the preceding expansion therefore yields the exact integer identity

$$u(x) = \sum_{\varnothing \neq \mathscr{A} \subseteq \mathscr{M}} (-2)^{|\mathscr{A}|-1} x^{\bigcup_{J\in\mathscr{A}} J}.$$

For any $U \subseteq [m]$, precisely $2^{m-|U|}$ inputs have $x_i = 1$ for all $i \in U$; hence $\sum_{x\in\mathbb{F}_2^m} x^U = 2^{m-|U|}$. Summing the last identity over all x now gives (3.1).

Suppose next that $m \geq 7$ and $\deg u \leq 3$, so $|J| \leq 3$ for all $J \in \mathscr{M}$. If $|\mathscr{A}| = 1$, its contribution to (3.1) is divisible by 2^{m-3}; if $|\mathscr{A}| = 2$, the union has at most six elements and the coefficient -2 supplies an additional factor of two, giving divisibility by 2^{m-5}. Both quantities are multiples of four when $m \geq 7$. For $|\mathscr{A}| \geq 3$, the coefficient $(-2)^{|\mathscr{A}|-1}$ itself is divisible by four. Hence $4 \mid \mathrm{wt}(u)$. Since $u(x) + a \cdot x$ also has degree at most three, the same divisibility applies to its weight, and

$$W_u(a) = 2^m - 2\,\mathrm{wt}(u + a \cdot x) \equiv 0 \pmod 8.$$

Finally, for $m \geq 4$ the parity of the weight is $\sum_x u(x) \pmod 2$. In the ANF, a monomial x^J contributes $2^{m-|J|}$ to this sum. As $|J| \leq 3 < m$, every contribution is even. Thus $\mathrm{wt}(u)$ is even. □

Lemma 3.2 (A shared residue and an affine next digit)**.** *Let $h : \mathbb{F}_2^9 \to \mathbb{F}_2$ have degree at most three and put $Q = W_h$. All $Q(a)$ have a common residue modulo 16, which is either 0 or 8. Moreover, for distinct i, j,*

$$16 \mid (Q(a + e_i) - Q(a)), \qquad 32 \mid \Delta_i\Delta_j Q(a). \tag{3.2}$$

Consequently $Q/16 \bmod 2$ is affine when $16 \mid Q$, and $(Q - 8)/16 \bmod 2$ is affine when $Q \equiv 8 \pmod{16}$.

Proof. For an integer-valued function T on V, define $\Delta_i T(a) = T(a + e_i) - T(a)$, where the subtraction is performed in $\mathbb{Z}$. By comparing the terms in $Q(a + e_i)$ and $Q(a)$,

$$\Delta_i Q(a) = -2 \sum_{x_i=1} (-1)^{h(x)+a\cdot x}.$$

Applying a second difference in a distinct direction e_j gives

$$\begin{aligned}\Delta_i\Delta_j Q(a) &= Q(a + e_i + e_j) - Q(a + e_i) - Q(a + e_j) + Q(a)\\ &= 4 \sum_{x_i=x_j=1} (-1)^{h(x)+a\cdot x}.\end{aligned}$$

Fixing one coordinate of h leaves a Boolean polynomial of degree at most three in eight variables; fixing two distinct coordinates leaves one in seven variables. The signs contributed by the fixed coordinates are immaterial to divisibility. Lemma 3.1 shows that the two sums on the right are divisible by eight. The factors -2 and 4 therefore give, respectively, $16 \mid \Delta_i Q(a)$ and $32 \mid \Delta_i \Delta_j Q(a)$.

Any two points of V can be joined by changes of one coordinate at a time. Since each such change alters Q by a multiple of 16, the value of $Q(a)$ is independent of a modulo 16. Lemma 3.1, applied in nine variables, also shows that every $Q(a)$ is divisible by 8. Its common residue modulo 16 is therefore some $r \in \{0, 8\}$.

Define the integer-valued function and its binary reduction by

$$T(a) = \frac{Q(a) - r}{16} \in \mathbb{Z}, \qquad b(a) = T(a) \bmod 2 \in \mathbb{F}_2.$$

The constant r disappears under a mixed difference. For distinct i, j, the divisibility proved above implies

$$\Delta_i \Delta_j T(a) = \frac{\Delta_i \Delta_j Q(a)}{16} \equiv 0 \pmod 2.$$

Reduction modulo two turns subtraction into addition. Therefore b satisfies

$$\begin{aligned} D_i D_j b(a) = {} & b(a + e_i + e_j) + b(a + e_i) \\ & + b(a + e_j) + b(a) = 0, \end{aligned} \qquad i \neq j,$$

where D_i denotes the usual Boolean derivative in direction e_i. To see what this implies, write the ANF of b as $b(a) = \sum_{S \subseteq [9]} b_S a^S$. For every squarefree monomial,

$$D_i D_j (a^S) = \begin{cases} a^{S \setminus \{i,j\}}, & \{i, j\} \subseteq S, \\ 0, & \text{otherwise.} \end{cases}$$

For fixed i, j, distinct supports containing $\{i, j\}$ give distinct monomials after removing those two indices, so their coefficients cannot cancel. If $b_S = 1$ for any $|S| \geq 2$, choose distinct $i, j \in S$; the coefficient of $a^{S \setminus \{i,j\}}$ in $D_i D_j b$ would be 1, contrary to the displayed identity. Thus $b_S = 0$ whenever $|S| \geq 2$, and $b(a) = b_\varnothing + \sum_{i=1}^{9} b_{\{i\}} a_i$ is affine. Taking $r = 0$ or $r = 8$ gives exactly the two conclusions of the lemma. □

Theorem 3.3 (Cubic spectral rigidity)**.** *For every nine-variable Boolean function h of degree at most three,*

$$\boxed{\|W_h\|_1 \leq 10240.} \tag{3.3}$$

If $W_h \equiv 8 \pmod{16}$, then the sharper bound $\|W_h\|_1 \le 10048$ holds. Equality in (3.3) is possible only with one of the following absolute spectra:

(3.4)

(A) $\#\{|W_h| = 16\} = 448, \quad \#\{|W_h| = 48\} = 64;$

(3.5)

(B) $\#\{|W_h| = 0\} = 64, \quad \#\{|W_h| = 16\} = 256, \quad \#\{|W_h| = 32\} = 192.$

Proof. Write $Q = W_h$. By Lemma 3.2, either $16 \mid Q$ or $Q \equiv 8 \pmod{16}$.

First assume $16 \mid Q$ and put $v = Q/16$. The parity $r = v \bmod 2$ is affine, so it is either constantly zero, constantly one, or balanced. Parseval gives

$$\sum_a v(a)^2 = 1024. \tag{3.6}$$

For even integers z and odd integers w, respectively, the exact inequalities

$$|z| \le \frac{z^2}{2}, \qquad |w| \le \frac{w^2+3}{4}, \qquad |w| \le \frac{w^2+1}{2} \tag{3.7}$$

follow by factoring the nonnegative differences. If $r = 0$, the first bound yields $\|Q\|_1 \le 8192$. If $r = 1$, the second yields

$$\|Q\|_1 \le 16\frac{1024 + 3 \cdot 512}{4} = 10240.$$

Equality requires $|v| \in \{1, 3\}$ pointwise. The energy equation then fixes 448 entries of magnitude one and 64 of magnitude three, which is spectrum (3.4).

If r is balanced, there are 256 odd positions. Apply the last inequality in (3.7) there and the first one on the remaining positions. We obtain

$$\sum_a |v(a)| \le \frac{1024 + 256}{2} = 640.$$

Equality forces the odd entries to be ± 1 and the even entries 0 or ± 2. The energy of the odd half is 256, leaving $768 = 192 \cdot 4$, so there are 192 even entries of magnitude two and 64 zeros. This is spectrum (3.5).

Finally suppose $Q \equiv 8 \pmod{16}$. The odd-valued function $u = Q/8$ has $\lambda = (u-1)/2 \bmod 2$ affine. Define $z(a) = (-1)^{\lambda(a)}u(a)$. Then $z \equiv 1 \pmod 4$ and $|z| = |u|$. Because $\lambda = c + b \cdot a$, Walsh inversion and Parseval give

$$\sum_a z(a) = 64(-1)^c s_h(b) \in \{-64, 64\}, \qquad \sum_a z(a)^2 = 4096. \tag{3.8}$$

For every integer $z \equiv 1 \pmod 4$,

$$16|z| \le 3z^2 - 2z + 15. \tag{3.9}$$

For $z \ge 1$ the difference of right and left sides is $3(z-1)(z-5)$; for $z \le -3$ it equals $(3z+5)(z+3)$. Both are nonnegative on the allowed progression. Summing and using (3.8) gives

$$\|Q\|_1 = 8\sum_a |z(a)| \le \frac{3 \cdot 4096 + 128 + 15 \cdot 512}{2} = 10048.$$

This is strictly smaller than 10240 and finishes the proof. □

4. Excluding the nonlinearity-243 case

4.1. **Parity and the support sum.** Every affine Boolean function on V has even weight. Thus all Hamming distances from f to affine functions have the parity of $\mathrm{wt}(f)$, and

$$\mathrm{NL}(f) \equiv \mathrm{wt}(f) \pmod 2. \tag{4.1}$$

The parity of $\mathrm{wt}(f)$ is also the coefficient of $x_1 \cdots x_9$ in the ANF. Consequently $\mathrm{NL}(f) = 243$ forces

$$\mathrm{wt}(f) \equiv 1 \pmod 2, \qquad \deg(f) = 9. \tag{4.2}$$

If $S = \mathrm{supp}(f)$ and $k_a = \#\{x \in S : a \cdot x = 1\}$, then $k_a \equiv a \cdot p(f) \pmod 2$. Expanding the character sum therefore gives the useful congruence

$$W_f(a) \equiv -2\,\mathrm{wt}(f) + 4(a \cdot p(f)) \pmod 8, \tag{4.3}$$

where $a \cdot p(f)$ is represented as 0 or 1 on the right.

Suppose $\mathrm{NL}(f) = 243$. Translating the input by $p(f)$ makes the support sum zero: since $\mathrm{wt}(f)$ is odd,

$$p(f(\cdot + p(f))) = p(f) + \mathrm{wt}(f)p(f) = 0.$$

The common Walsh residue modulo eight is then 2 or 6; output complementation, if necessary, changes it to 2. These operations preserve nonlinearity and the absolute Walsh spectrum. We may therefore assume throughout the next two subsections that

$$p(f) = 0, \qquad W_f(a) \equiv 2 \pmod 8, \qquad \|W_f\|_\infty = 26. \tag{4.4}$$

No assumption about the unknown sign $s_f(0)$ is made.

4.2. **A rigid quadratic residue.** From (4.4),

$$W_f(a) \in \{-22, -14, -6, 2, 10, 18, 26\}.$$

Put $v = (W_f - 2)/8$, so that $v(a) \in \{-3, -2, -1, 0, 1, 2, 3\}$, and set $q = v \bmod 2$. Lemma 2.1 gives $\deg q \le 2$. Writing $\sigma = s_f(0) \in \{1, -1\}$, inversion at zero and Parseval produce the exact moment equations

$$\sum_a v(a) = 64\sigma - 128, \qquad \sum_a v(a)^2 = 4128 - 32\sigma \ge 4096. \tag{4.5}$$

Let $m = \mathrm{wt}(q)$. On a zero of q, v is even and $v^2 \le 4$; on a one, v is odd and $v^2 \le 9$. Hence

$$4096 \le \sum_a v(a)^2 \le 4(512 - m) + 9m = 2048 + 5m, \qquad m \ge 410. \tag{4.6}$$

If q were not identically one, $1 + q$ would be a nonzero Boolean function of degree at most two. Lemma 2.2 would give $\mathrm{wt}(1 + q) \ge 128$, or $m \le 384$, a contradiction. We have proved

$$q \equiv 1, \quad W_f(a) \equiv 10 \pmod{16}, \quad W_f(a) \in \{-22, -6, 10, 26\}. \tag{4.7}$$

4.3. **Determining the sign at the origin.** The decisive point is that the normalization actually forces $s_f(0) = 1$. Suppose, to the contrary, that $\sigma = -1$. Let A, B, C, D be the respective multiplicities of the four values $-22, -6, 10, 26$ in (4.7). Counting, inversion at the origin, and Parseval give

$$\begin{aligned} A + B + C + D &= 512, \\ -22A - 6B + 10C + 26D &= -512, \\ 484A + 36B + 100C + 676D &= 262144. \end{aligned}$$

Elementary elimination yields

$$A + D = 456, \qquad B = 808 - 3A, \quad C = 3A - 752, \quad 251 \le A \le 269. \tag{4.8}$$

Define a Boolean phase on frequency space by the *integer* quotient

$$h(a) = \frac{W_f(a) - 10}{16} \pmod 2. \tag{4.9}$$

Lemma 2.1 gives $\deg h \le 3$. Here the *sign of the phase* at a frequency a means $s_h(a) = (-1)^{h(a)} \in \{1, -1\}$; it is not, in general, the ordinary sign of the integer $W_f(a)$. To determine $s_h(a)$, first evaluate the integer quotient in (4.9) and then reduce it modulo two. For the four possible Walsh values, this gives

$W_f(a)$	-22	-6	10	26
$(W_f(a) - 10)/16$	-2	-1	0	1
$h(a)$	0	1	0	1
$s_h(a) = (-1)^{h(a)}$	1	-1	1	-1

Indeed, the quotients -2 and 0 are even, so the corresponding phase values are zero and their signs are 1. The quotients -1 and 1 are odd, so the phase values are one and their signs are -1. This also illustrates why reduction modulo two must be performed *after* the integer division. Thus

$$\mathrm{wt}(h) = B + D = 1264 - 4A, \tag{4.10}$$

$$W_h(0) = 8A - 2016, \tag{4.11}$$

$$\sum_a W_f(a) s_h(a) = 16A - 14528. \tag{4.12}$$

Using (2.4) and the triangle inequality,

$$\|W_h\|_1 \ge 14528 - 16A. \tag{4.13}$$

Theorem 3.3 forces $A \ge 268$, leaving only two integer values. If $A = 268$, the right side of (4.13) is 10240 and equality in the cubic norm theorem is forced. But (4.11) gives $W_h(0) = 128$, whereas the two possible equality spectra have every magnitude at most 48. If $A = 269$, equation (4.11) gives $W_h(0) = 136 \equiv 8 \pmod{16}$. By Lemma 3.2, all Walsh coefficients of h lie in this residue class. The sharper bound $\|W_h\|_1 \le 10048$ contradicts the lower bound 10224 from (4.13). Both possibilities are impossible. Therefore

$$\boxed{s_f(0) = 1, \qquad f(0) = 0.} \tag{4.14}$$

4.4. **The flip theorem and its converse.** For the normalized f, change the output at the origin: $g = f + \delta_0$. By (2.5),

$$W_g(a) = W_f(a) - 2 \in \{-24, -8, 8, 24\}.$$

Parseval rules out the possibility that every coefficient has magnitude eight. Thus $\|W_g\|_\infty = 24$ and $\mathrm{NL}(g) = 244$.

We now transfer this conclusion to the function before normalization. Write f_0 for the original function of nonlinearity 243, and set $p = p(f_0)$. As in the normalization preceding (4.4), we chose $\beta \in \mathbb{F}_2$ so that

$$\widetilde{f}(x) = f_0(x + p) + \beta$$

has the required Walsh residue. The normalized function f used above is precisely $\widetilde{f}$. Consequently, the function obtained by flipping its value at the origin is

$$\begin{aligned}\widetilde{g}(x) &= \widetilde{f}(x) + \delta_0(x) \\ &= f_0(x + p) + \beta + \delta_0(x) \\ &= (f_0 + \delta_p)(x + p) + \beta.\end{aligned}$$

The last equality follows from $\delta_0(x) = \delta_p(x + p)$: each indicator equals 1 exactly when $x = 0$. Equivalently, putting $y = x + p$ and using $p + p = 0$ in $\mathbb{F}_2^9$, we have

$$f_0(y) + \delta_p(y) = \widetilde{g}(y + p) + \beta.$$

Input translation and addition of a constant only multiply Walsh coefficients by signs. More explicitly,

$$W_{\widetilde{g}}(a) = (-1)^{a \cdot p + \beta} W_{f_0 + \delta_p}(a), \qquad a \in V.$$

Their maximum absolute Walsh coefficients, and hence their nonlinearities, are equal. Since $\mathrm{NL}(\widetilde{g}) = 244$, we obtain

$$\boxed{\mathrm{NL}(f_0 + \delta_{p(f_0)}) = 244}.$$

Thus the improvement at the normalized origin corresponds exactly to flipping the original function at its support sum.

Conversely, suppose $\mathrm{NL}(g) = 244$. Then $\mathrm{wt}(g)$ is even: otherwise $W_g(a) \equiv 2 \pmod 4$ for every a, which would imply $|W_g(a)| \le 22$ and contradict Parseval. Hence any $g + \delta_t$ has odd weight, and thus odd nonlinearity by (4.1). The one-bit bound and the universal upper bound 244 show that

$$243 \le \mathrm{NL}(g + \delta_t) \le 244, \qquad \mathrm{NL}(g + \delta_t) \text{ odd};$$

therefore $\mathrm{NL}(g + \delta_t) = 243$ for every t. We will prove in Proposition 5.1 that $W_g(a) \equiv 8 \pmod{16}$. Formula (4.3) then shows that $p(g) = 0$. Finally

$$p(u + \delta_t) = p(u) + t$$

for every Boolean function u, because toggling the point t changes the support sum by t. If $f + \delta_t$ has nonlinearity 244, then $0 = p(f + \delta_t) = p(f) + t$, so

$t = p(f)$. This proves all assertions of Theorem 1.2, including the bijection and the identity

$$\big|\{f : \mathrm{NL}(f) = 243\}\big| = 512\big|\{g : \mathrm{NL}(g) = 244\}\big|. \tag{4.15}$$

5. The hypothetical spectrum of nonlinearity 244

From now on assume for contradiction that $\mathrm{NL}(f) = 244$, or equivalently $\|W_f\|_\infty = 24$.

Proposition 5.1 (Four-value Walsh rigidity)**.** *Every Walsh coefficient of f is* 8 *modulo* 16*, and*

$$W_f(a) \in \{-24, -8, 8, 24\}, \qquad \#\{|W_f| = 8\} = 64, \quad \#\{|W_f| = 24\} = 448. \tag{5.1}$$

Furthermore $4 \mid \mathrm{wt}(f)$ *and* $p(f) = 0$.

Proof. As noted above, $\mathrm{wt}(f)$ must be even, so $4 \mid W_f(a)$ for all a. Formula (2.7) shows that $W_f/4 \bmod 2$ is affine. If this affine function were nonzero, there would be at least 256 frequencies with $W_f \equiv 4 \pmod 8$ and therefore $|W_f| \le 20$; the remaining frequencies have $|W_f| \le 24$. But then

$$\sum_a W_f(a)^2 \le 256 \cdot 20^2 + 256 \cdot 24^2 = 249856 < 262144,$$

contrary to Parseval. Thus $8 \mid W_f$.

By Lemma 2.1, $q = W_f/8 \bmod 2$ has degree at most two. If $q \neq 1$, Lemma 2.2 applied to $1+q$ provides at least 128 frequencies at which $16 \mid W_f$ and $|W_f| \le 16$. Hence

$$\sum_a W_f(a)^2 \le 128 \cdot 16^2 + 384 \cdot 24^2 = 253952 < 262144,$$

another contradiction. Therefore $q = 1$, giving the four permitted values. If k entries have magnitude 24, Parseval says $576k + 64(512 - k) = 262144$, whence $k = 448$.

Finally, $W_f \equiv 0 \pmod 8$ and (4.3) imply $4(a \cdot p(f)) \equiv 0 \pmod 8$ for every a. Thus $p(f) = 0$; taking $a = 0$ in the same formula gives $4 \mid \mathrm{wt}(f)$. □

Lemma 5.2 (A degree-six/cubic complement relation)**.** *There is a cubic Boolean function*

$$h(a) = \frac{W_f(a) - 8}{16} \pmod 2 \tag{5.2}$$

for which

$$\deg(f) \le 6, \qquad h_I = f_{I^c} \quad \text{for every } I \subseteq [9],\ |I| = 3. \tag{5.3}$$

Here subscripts denote ANF coefficients. Moreover

$$\sum_a s_h(a) W_f(a) = -10240, \qquad \|W_h\|_1 = 10240, \quad \deg(h) = 3. \tag{5.4}$$

Proof. Let $L = v + U$ be an arbitrary affine seven-flat, so $U^\perp$ has four elements. Character orthogonality gives

$$\sum_{x\in L} s_f(x) = \frac{1}{4}\sum_{a\in U^\perp} (-1)^{a\cdot v} W_f(a).$$

Each $W_f(a) \equiv 8 \pmod{16}$, and the four character signs sum to 0 or 4. The integer on the right is therefore divisible by four. As the left side equals $128 - 2\,\mathrm{wt}(f|_L)$, the weight of f on every affine seven-flat is even. Every seventh Boolean derivative vanishes, proving $\deg f \le 6$.

The quotient in (5.2) is integer-valued, and Lemma 2.1 gives $\deg h \le 3$. For $|I| = 3$, (2.7) shows that the coefficient of a^I in $(W_f - 8)/16$ equals $-32 + M_I(f)$; hence $h_I = M_I(f) \pmod 2$. The parity of a face weight is the coefficient of the full monomial in its six free coordinates. Since $\deg f \le 6$, this is exactly f_{I^c}, establishing (5.3).

For $W_f = \pm 8$ the signed product $s_h W_f$ is $+8$; for $W_f = \pm 24$ it is -24. Using (5.1),

$$\sum_a s_h(a) W_f(a) = 64 \cdot 8 - 448 \cdot 24 = -10240.$$

By (2.4), $\|W_h\|_1 \ge 10240$; Theorem 3.3 makes this an equality. If $\deg h \le 2$, equation (2.8) would instead give $\|W_h\|_1 \le 8192$. Thus $\deg h = 3$. □

Theorem 3.3 now leaves exactly spectra (3.4) and (3.5). We first eliminate the second spectrum without any cubic classification.

6. Eliminating spectrum (B) by hyperplane slicing

The two spectra in Theorem 3.3 arise from an estimate for the cubic phase h alone. We now exploit the fact that h is associated with the original function f through (5.2). In particular, the Walsh coefficients of f are constrained by Parseval's identity on each of two complementary hyperplanes. These additional constraints will exclude spectrum (3.5).

6.1. The two hyperplane slices and their parity.

Fix a nonzero vector $e \in V = \mathbb{F}_2^9$. The subspace generated by e is $\langle e\rangle = \{0, e\}$, and its cosets are the 256 disjoint pairs $\{u, u + e\}$ with $u \in V$. Choose a set $R \subseteq V$ that contains exactly one element of each such pair. Thus $|R| = 256$, and whenever we sum over u below we mean $u \in R$. The particular choice of a representative will not affect any of the absolute values, squares, or parities used in the argument.

Consider the two affine hyperplanes in the *input* space,

$$H_0 = \{x \in V : e \cdot x = 0\}, \qquad H_1 = \{x \in V : e \cdot x = 1\}.$$

Since $e \neq 0$, each hyperplane has 256 points. For $j \in \{0, 1\}$ and $u \in V$, put

$$S_j(u) = \sum_{x\in H_j} (-1)^{f(x)+u\cdot x}.$$

The defining Walsh sum splits into contributions from H_0 and H_1:

$$W_f(u) = S_0(u) + S_1(u).$$

Moreover, on H_j the factor $(-1)^{e\cdot x}$ is constantly $(-1)^j$, so

$$W_f(u+e) = S_0(u) - S_1(u).$$

Recall from Proposition 5.1 that $v_f = W_f/8$ takes its values in $\{-3,-1,1,3\}$. Solving the last two equations for S_0, S_1 gives

$$\begin{aligned} A_e(u) &= \frac{v_f(u)+v_f(u+e)}{2} = \frac{S_0(u)}{8}, \\ B_e(u) &= \frac{v_f(u)-v_f(u+e)}{2} = \frac{S_1(u)}{8}. \end{aligned} \tag{6.1}$$

Both are integers, since $v_f(u)$ and $v_f(u+e)$ are odd. Replacing u by $u+e$ leaves $A_e(u)$ unchanged and reverses the sign of $B_e(u)$. Hence the quantities relevant to us depend only on the coset $u + \langle e\rangle$.

We next justify the two Parseval identities used below. The characters $x \mapsto (-1)^{u\cdot x}$ restricted to the linear hyperplane H_0 depend only on $u + \langle e\rangle$, and the 256 cosets give all characters of $H_0 \cong \mathbb{F}_2^8$. Thus the numbers $S_0(u)$, $u \in R$, form the unnormalised Walsh transform of $f|_{H_0}$. Choose any $t \in H_1$ and write $H_1 = t + H_0$. Then

$$S_1(u) = (-1)^{u\cdot t} \sum_{y\in H_0} (-1)^{f(t+y)+u\cdot y},$$

so $S_1(u)$ differs by a sign from the Walsh coefficient of the translated restriction $y \mapsto f(t+y)$ on H_0. Eight-variable Parseval, applied separately to the two restrictions, therefore gives $\sum_{u\in R} S_j(u)^2 = 256^2$ for $j = 0, 1$. Dividing by 8^2 yields

$$\sum_{u\in R} A_e(u)^2 = \sum_{u\in R} B_e(u)^2 = 1024. \tag{6.2}$$

We now determine the parity of $A_e(u)$. By the definition of the cubic phase in (5.2),

$$v_f(a) \equiv 1 + 2h(a) \equiv (-1)^{h(a)} \pmod 4, \qquad a \in V.$$

Write $\alpha = h(u)$ and $\beta = h(u+e)$, treating these Boolean values as the ordinary integers 0 or 1. Since $(-1)^\alpha \equiv 1 - 2\alpha \pmod 4$ and similarly for β,

$$v_f(u) + v_f(u+e) \equiv (1-2\alpha) + (1-2\beta) = 2 - 2(\alpha+\beta) \pmod 4.$$

The left side is even, so dividing this congruence by 2 gives a congruence modulo 2. Using (6.1), we obtain

$$A_e(u) \bmod 2 = 1 + \alpha + \beta = 1 + h(u) + h(u+e) =: q_e(u), \tag{6.3}$$

where the final additions are in $\mathbb{F}_2$. In particular, $A_e(u)$ is odd precisely when $h(u) = h(u+e)$; if the two phase values differ, $A_e(u)$ is even.

The Boolean derivative $D_e h(u) = h(u) + h(u+e)$ satisfies $D_e h(u+e) = D_e h(u)$, since $e + e = 0$. Therefore $q_e = 1 + D_e h$ is constant on each coset of

$\langle e\rangle$ and defines a Boolean function on $V/\langle e\rangle$. Because $\deg h \le 3$, its derivative $D_e h$ has degree at most two (for example, choose linear coordinates with e as one basis vector). Thus q_e is a quadratic Boolean function on the quotient. Finally, $v_f(u) = A_e(u) + B_e(u)$ is odd, so the two integers $A_e(u), B_e(u)$ always have opposite parity. In particular,

$$B_e(u) \bmod 2 = 1 + q_e(u) = D_e h(u). \tag{6.4}$$

6.2. **A uniform bound for the quotient weight.** Let $m = \mathrm{wt}(q_e)$, where the weight is computed on the 256 cosets, not on the original 512-point space. Each of $v_f(u), v_f(u+e)$ belongs to $\{\pm 1, \pm 3\}$. Consequently, by (6.1), the pair of magnitudes $(|A_e(u)|, |B_e(u)|)$ must be one of

$$(3,0),\ (1,2),\ (1,0),\ (0,3),\ (2,1),\ (0,1).$$

Denote by N_{ij} the number of cosets for which $(|A_e|, |B_e|) = (i, j)$. The six possibilities, together with the parity of A_e, are

$(\lvert A_e\rvert, \lvert B_e\rvert)$	$(3,0)$	$(1,2)$	$(1,0)$	$(0,3)$	$(2,1)$	$(0,1)$
multiplicity	N_{30}	N_{12}	N_{10}	N_{03}	N_{21}	N_{01}
q_e	1	1	1	0	0	0

In particular,

$$N_{30} + N_{12} + N_{10} = m, \qquad N_{03} + N_{21} + N_{01} = 256 - m.$$

Substituting the six pairs into the two energy identities (6.2) gives, respectively,

$$9N_{30} + N_{12} + N_{10} + 4N_{21} = 1024,$$
$$4N_{12} + 9N_{03} + N_{21} + N_{01} = 1024.$$

Using the preceding two counting equations, we may rewrite these as

$$8N_{30} + m + 4N_{21} = 1024, \qquad 8N_{03} + 256 - m + 4N_{12} = 1024. \tag{6.5}$$

In particular, the first equality implies $m \equiv 0 \pmod 4$.

There is a third counting constraint. Recall that exactly 448 of the 512 values $v_f(a)$ have magnitude 3 by (5.1). A pair of type $(3,0)$ or $(0,3)$ comes from two values of magnitude 3, while a pair of type $(1,2)$ or $(2,1)$ comes from one value of magnitude 3 and one of magnitude 1. The remaining two types use only values of magnitude 1. Counting the 448 large-magnitude values therefore gives

$$2(N_{30} + N_{03}) + N_{12} + N_{21} = 448. \tag{6.6}$$

Put $r = N_{30} + N_{03}$. Since the total number of cosets is 256,

$$N_{12} + N_{21} \le 256 - r.$$

Together with (6.6), this yields $448 \le 2r + (256 - r) = 256 + r$, hence $r \ge 192$. On the other hand, $N_{30} \le m$, and the second equation in (6.5) implies

$$8N_{03} \le 1024 - (256 - m) = 768 + m, \qquad N_{03} \le 96 + \frac{m}{8}.$$

Therefore

$$192 \le r = N_{30} + N_{03} \le m + 96 + \frac{m}{8},$$

which is equivalent to $m \ge 256/3$. Since $4 \mid m$, we obtain $m \ge 88$.

It remains to obtain the upper bound. The preceding argument treats the two hyperplane slices symmetrically: both have the same energy 1024, and the six possible pairs remain the same after exchanging A_e and B_e (the types $(3,0)$ and $(0,3)$, $(1,2)$ and $(2,1)$, and $(1,0)$ and $(0,1)$ are simply exchanged). If we regard B_e as the first slice, its parity function is $q'_e = B_e \bmod 2 = 1 + q_e$ by (6.4). Thus its weight on the quotient is

$$\mathrm{wt}(q'_e) = 256 - \mathrm{wt}(q_e) = 256 - m.$$

Applying exactly the same counting argument to the slices in this reversed order gives $256 - m \ge 88$, or $m \le 168$. We have proved the bound, valid for *every* nonzero e,

$$\boxed{88 \le \mathrm{wt}(q_e) \le 168 \qquad (e \ne 0).} \tag{6.7}$$

Proposition 6.1. *Spectrum* (3.5) *is impossible for the cubic phase h attached to a nonlinearity-*244 *function.*

Proof. Assume that W_h has spectrum (3.5), and write $Q = W_h$. By the equality analysis in Theorem 3.3, the parity function

$$\ell(a) = \frac{Q(a)}{16} \bmod 2$$

is a nonconstant affine Boolean function. In particular it is balanced, taking each value 0 and 1 at exactly 256 frequencies. There consequently exist a *nonzero* vector $e \in V$ and a constant $c \in \mathbb{F}_2$ such that

$$\ell(a) = a \cdot e + c.$$

We use this e in the hyperplane-slice bound (6.7), which was established for every nonzero direction.

The equality conditions for spectrum (3.5) give more information than the three multiplicities alone. Where $\ell(a) = 1$, the integer $Q(a)/16$ is odd, and equality requires $|Q(a)/16| = 1$; hence $|Q(a)| = 16$ at all 256 such frequencies. Where $\ell(a) = 0$, the quotient is even, and equality requires its magnitude to be 0 or 2. Of those 256 frequencies, 192 have $|Q(a)| = 32$ and the remaining 64 have $Q(a) = 0$. It follows that the squared Walsh masses on these two halves are

$$M_1 := \sum_{\ell(a)=1} Q(a)^2 = 256 \cdot 16^2 = 65536,$$

$$M_0 := \sum_{\ell(a)=0} Q(a)^2 = 192 \cdot 32^2 = 196608.$$

We next convert this imbalance into a statement about a derivative of h. The Wiener–Khinchin identity in our unnormalised convention is

$$\mathrm{AC}_h(e) := \sum_{x\in V}(-1)^{h(x)+h(x+e)} = \frac{1}{512}\sum_{a\in V}(-1)^{a\cdot e}Q(a)^2. \tag{6.8}$$

For completeness, this identity follows by expanding both factors of $Q(a)^2$ as Walsh sums and applying character orthogonality:

$$\frac{1}{512}\sum_{a\in V}(-1)^{a\cdot(e+x+y)} = \begin{cases}1, & y = x+e,\\ 0, & y \neq x+e.\end{cases}$$

Because $\ell(a) = a\cdot e + c$, we have $(-1)^{a\cdot e} = (-1)^c(-1)^{\ell(a)}$. The terms with $\ell = 0$ and $\ell = 1$ therefore enter (6.8) with opposite signs. Substituting the two masses gives

$$\mathrm{AC}_h(e) = \frac{(-1)^c}{512}(M_0 - M_1) = \frac{(-1)^c}{512}(196608 - 65536) = (-1)^c 256.$$

In particular,

$$|\,\mathrm{AC}_h(e)| = 256. \tag{6.9}$$

Now consider the Boolean derivative $D_e h(x) = h(x) + h(x+e)$. Since $e + e = 0$,

$$D_e h(x+e) = h(x+e) + h(x) = D_e h(x).$$

Thus its two values on each coset $\{u, u+e\}$ are identical. Choose one representative from each of the 256 cosets, as at the start of this section. Each coset contributes twice the same sign to the autocorrelation sum, and hence

$$\mathrm{AC}_h(e) = 2\sum_{u\in R}(-1)^{D_e h(u)}.$$

By (6.9),

$$\left|\sum_{u\in R}(-1)^{D_e h(u)}\right| = 128. \tag{6.10}$$

Let k be the number of cosets on which $D_e h = 1$. The other $256 - k$ cosets have derivative zero, so the sum in (6.10) equals

$$(256 - k) - k = 256 - 2k.$$

It follows that $|256 - 2k| = 128$, and therefore $k = 64$ or $k = 192$. Finally, $q_e = 1 + D_e h$ by (6.3), so q_e is the complement of the derivative on the quotient. Its weight is

$$\mathrm{wt}(q_e) = 256 - k \in \{64, 192\}.$$

Neither value lies in the interval $[88, 168]$ required by (6.7). This contradiction excludes spectrum (3.5). □

7. A double-pairing obstruction in spectrum (A)

The sole remaining spectrum is

$$W_h(x) \in \{-48, -16, 16, 48\}, \qquad \#\{|W_h| = 48\} = 64. \tag{7.1}$$

Equality in (2.4) and (5.4) fixes the sign of f at *every* input:

$$\boxed{s_f(x) = -\operatorname{sgn} W_h(x).} \tag{7.2}$$

Let $\omega = (\omega_I)_{|I|=3}$ be the homogeneous cubic top part of h. For any three-element set I, define

$$(\omega^{\#})_I = \sum_{\substack{\{J,K\} \text{ unordered} \\ J \dot\cup K = I^c \\ |J|=|K|=3}} \omega_J \omega_K \quad \in \mathbb{F}_2, \qquad \omega^{\#\#} = (\omega^{\#})^{\#}. \tag{7.3}$$

Each sum has ten possible unordered partitions.

Proposition 7.1 (Double-complement-pairing identity)**.** *The cubic top part of a hypothetical nonlinearity-*244 *function satisfies*

$$\boxed{\omega^{\#\#} = \omega.} \tag{7.4}$$

Proof. Put $v = W_h/16$, an odd integer-valued function with values $\pm 1, \pm 3$. Let d_S be the integer coefficient of a^S in its multilinear representation. For nonempty S, with $s = |S|$, formula (2.7) gives

$$d_S = (-1)^s 32 - (-1)^s 2^{s-3} M_S(h). \tag{7.5}$$

We next justify carefully why the coefficients d_S are even whenever S is nonempty. For $T \subseteq [9]$, let $\mathbf{1}_T \in V$ denote the vector with ones precisely in the positions belonging to T. Evaluating the multilinear representation of v at this vector gives

$$v(\mathbf{1}_T) = \sum_{R \subseteq T} d_R.$$

Möbius inversion, as in (2.6), therefore yields

$$d_S = \sum_{T \subseteq S} (-1)^{|S|-|T|} v(\mathbf{1}_T). \tag{7.6}$$

Since v takes only odd integer values, for each T there is an integer k_T such that $v(\mathbf{1}_T) = 1 + 2k_T$. Substitution in (7.6) shows that, for every nonempty S,

$$\begin{aligned} d_S &= \sum_{T\subseteq S} (-1)^{|S|-|T|} + 2\sum_{T\subseteq S} (-1)^{|S|-|T|} k_T \\ &= (1-1)^{|S|} + 2\sum_{T\subseteq S} (-1)^{|S|-|T|} k_T \\ &= 2\sum_{T\subseteq S} (-1)^{|S|-|T|} k_T. \end{aligned}$$

Thus $2 \mid d_S$ whenever $S \neq \varnothing$; by contrast, the constant coefficient is $d_\varnothing = v(0)$, which is odd. Notice that this argument uses the parity of the *values* of v and does not require any divisibility of the individual face weights in (7.5).

For clarity, when $s = 1$ and $s = 2$ the coefficient formula (7.5) reads, respectively,

$$d_{\{i\}} = -32 + \frac{M_{\{i\}}(h)}{4}, \qquad d_{\{i,j\}} = 32 - \frac{M_{\{i,j\}}(h)}{2}.$$

Both expressions are integers because they are coefficients of the integer-valued function v. Their evenness, just proved, also implies $8 \mid M_{\{i\}}(h)$ and $4 \mid M_{\{i,j\}}(h)$. For $s = 3$, the six-dimensional face weight is even; for $s = 4$ and $s = 5$, parity of the five- and four-dimensional face weights in Lemma 3.1 gives, together with (7.5),

$$2 \mid d_S \ (1 \le s \le 3), \qquad 4 \mid d_S \ (s = 4), \qquad 8 \mid d_S \ (s \ge 5). \tag{7.7}$$

(The last assertion for $s \ge 6$ follows directly from (7.5).)

For $|I| = 3$, set $b_I = d_I/2 = -16 + M_I(h)/2$. Apply (3.1) to the six-variable face $x_i = 1$ for $i \in I$, now reducing its weight modulo four. Single monomials contribute multiples of eight; terms involving three or more selected monomials contribute multiples of four. Two selected monomials contribute modulo four only when they are disjoint cubics whose union is the six free variables; their coefficient is -2. Terms of lower degree, including those induced by fixing the other three coordinates, cannot participate in such a partition. We conclude that

$$b_I \equiv (\omega^{\#})_I \pmod 2. \tag{7.8}$$

We now extract the degree-six coefficients of the forced Boolean function without dividing any expression in $\mathbb{F}_2$. For a signed integer z define its jth binary digit by $\mathrm{bit}_j(z) = \lfloor z/2^j \rfloor \bmod 2$. This definition also works for negative integers. Because $v \in \{\pm 1, \pm 3\}$, the digit $\mathrm{bit}_2(v)$ is one exactly when v is negative. Relation (7.2) therefore says

$$f = 1 + \mathrm{bit}_2(v) \quad \text{in } \mathbb{F}_2. \tag{7.9}$$

Write $v = c + 2t$, where $c = v(0) = 2k + 1$ is odd and $t = (v - c)/2$ is an integer-valued polynomial with no constant term.

The elementary binary-carry identity

$$\mathrm{bit}_2(2(k+t)+1) = \mathrm{bit}_1(t) + (k \bmod 2)\,\mathrm{bit}_0(t) + \mathrm{bit}_1(k) \pmod 2$$

shows that every degree-six coefficient of f comes from $\mathrm{bit}_1(t)$, since $\mathrm{bit}_0(t)$ has degree at most three by (7.7).

We next explain the binary carry that occurs when taking the second digit of an integer polynomial. Let $t(a) = \sum_{S\subseteq[9]} b_S a^S$ have integer coefficients, where $a^S = \prod_{i\in S} a_i$ and $a^\varnothing = 1$. For each S, use Euclidean division by two to write

$$b_S = 2q_S + r_S, \qquad q_S = \left\lfloor \frac{b_S}{2} \right\rfloor \in \mathbb{Z}, \qquad r_S \in \{0,1\}.$$

This decomposition is valid even for negative coefficients; for example, $-1 = 2(-1)+1$. Since a^S equals either zero or one on the Boolean cube, at any fixed input a we have

$$t(a) = 2U(a) + R(a), \qquad U(a) = \sum_S q_S a^S, \qquad R(a) = \sum_S r_S a^S.$$

Here $U(a)$ is an arbitrary integer, while $R(a)$ is the *number* of indices S for which both $r_S = 1$ and $a^S = 1$. In particular, $R(a)$ is nonnegative. Taking the integer quotient by two and reducing it modulo two yields

$$\mathrm{bit}_1(t(a)) \equiv U(a) + \left\lfloor \frac{R(a)}{2} \right\rfloor \pmod 2. \tag{7.10}$$

For every nonnegative integer r, one has

$$\left\lfloor \frac{r}{2} \right\rfloor \equiv \binom{r}{2} \pmod 2. \tag{7.11}$$

Indeed, if $r = 2m$ then $\binom{r}{2} = m(2m-1) \equiv m \pmod 2$, and if $r = 2m+1$ then $\binom{r}{2} = m(2m+1) \equiv m \pmod 2$; in both cases $m = \lfloor r/2 \rfloor$. The binomial coefficient $\binom{R(a)}{2}$ counts unordered pairs among the $R(a)$ active indices. Hence, with any fixed ordering of the subsets $S \subseteq [9]$,

$$\binom{R(a)}{2} = \sum_{S<T} (r_S a^S)(r_T a^T) = \sum_{S<T} r_S r_T a^{S\cup T}.$$

The last equality uses the Boolean-input identity $a^S a^T = a^{S\cup T}$, including when S and T intersect. Substituting this expression into (7.10), and recalling that $q_S = \lfloor b_S/2 \rfloor$ and $r_S \equiv b_S \pmod 2$, we obtain the exact Boolean expansion

$$\mathrm{bit}_1(t) = \sum_S \left\lfloor \frac{b_S}{2} \right\rfloor a^S + \sum_{S<T} (b_S \bmod 2)(b_T \bmod 2) a^{S\cup T} \pmod 2. \tag{7.12}$$

The derivation uses only integer division and counting of active terms; in particular, no positivity assumption on the coefficients b_S is needed.

By (7.7), the coefficients $b_S = d_S/2$ are even for $|S| \geq 4$ and divisible by four for $|S| \geq 5$. The first sum in (7.12) therefore has degree at most

four. Only supports of size at most three can occur in the second sum; their union can have size six only when both are disjoint triples. Combining this with (7.8), for $|L| = 6$ we get

$$f_L = \sum_{\substack{J\dot\cup K=L\\ |J|=|K|=3}}^{\text{unordered}} b_J b_K = \sum_{J\dot\cup K=L}^{\text{unordered}} (\omega^{\#})_J (\omega^{\#})_K = (\omega^{\#\#})_{L^c}.$$

Finally, Lemma 5.2 gives $f_L = h_{L^c} = \omega_{L^c}$, proving (7.4). □

Lemma 7.2 (A constant cubic residue)**.** *Define the cubic partition invariant*

$$\Pi(\omega) = \sum_{\substack{\{I,J,K\}\ \text{unordered}\\ I\dot\cup J\dot\cup K=[9]\\ |I|=|J|=|K|=3}} \omega_I\omega_J\omega_K \in \mathbb{F}_2. \tag{7.13}$$

Every nine-variable Boolean cubic h with cubic top part ω satisfies

$$\operatorname{wt}(h) \equiv 4\Pi(\omega) \pmod 8, \qquad W_h(a) \equiv 8\Pi(\omega) \pmod{16}.$$

Consequently the phase in spectrum (7.1) *satisfies* $\Pi(\omega) = 0$.

Proof. Apply the weight expansion (3.1) in nine variables modulo eight. One or two monomials contribute zero; four or more contribute zero because of their powers of two. Three monomials contribute only if they cover all nine variables, which forces three disjoint cubic supports. Their integer factor is four. The weight congruence follows. The Walsh residue is uniform by Lemma 3.2, and at the origin equals $512 - 2\operatorname{wt}(h) \equiv 8\Pi(\omega) \pmod{16}$. Spectrum (7.1) has residue zero modulo 16. □

8. Classification, normal forms, and finite completion

8.1. **The cubic classification and its exact filter.** The cubic top part corresponds, by taking third derivatives, to an alternating trilinear form on V. The classification of Brier and Langevin [1] and the independently formulated classification of Hora and Pudlák [4] give 348 nonzero $\mathrm{GL}(9,2)$-orbits, with the zero form making 349 in total. The public Langevin representative list [7] uses indices $1,\dots,349$, with 349 the zero cubic. The audit reported in [3] computes 349 distinct orbit signatures on its embedded copy of the catalogue (including zero). Together with the published orbit count, this certifies coverage; the number of lines alone would not suffice.

Here normalization by $L \in \mathrm{GL}(9,2)$ is legitimate for the *pair* (f,h): replace

$$h'(a) = h(La), \qquad f'(x) = f(L^{-T}x). \tag{8.1}$$

Their Walsh transforms satisfy $W_{f'}(a) = W_f(La)$ and $W_{h'}(x) = W_h(L^{-T}x)$. Thus all spectral and sign relations are preserved, including the complement identity in the new standard coordinates. We may apply Proposition 7.1 *after* normalization; no separate covariance assertion for # is being assumed.

Finite certificate 8.1 (Exact cubic catalogue filter)**.** On the 349 indexed representatives,

$$\{j : \omega_j^{\#\#} = \omega_j,\ \Pi(\omega_j) = 0\} = \{282, 290, 349\}. \tag{8.2}$$

For comparison, the fixed forms with $\Pi = 1$ have indices $320, 326, 329$. The calculation is a finite evaluation of (7.3) and (7.13) on the complete public representative list, not an orbit-invariant assumption. The nonzero surviving representatives are

$$c_{282} = 125 + 145 + 345 + 346 + 137 + 248 + 349 + 569 + 789, \tag{8.3}$$

$$\begin{aligned} c_{290} = {} & 125 + 235 + 245 + 345 + 136 + 127 + 347 + 128 \\ & + 238 + 148 + 129 + 349 + 569 + 789. \end{aligned} \tag{8.4}$$

Here ijk abbreviates the monomial $a_i a_j a_k$ and all additions are in $\mathbb{F}_2$.

The accompanying catalogue-filter program implements this check on the public notebook or original text catalogue. For every three-set I, enumerate the ten unordered partitions of I^c into two triples, compute $\omega^{\#}$, and repeat; enumerate the 280 partitions of $[9]$ into three unordered triples to compute Π. These calculations return exactly (8.2) when applied to the published embedded 349-record list. Since h has degree three by Lemma 5.2, the zero form 349 is excluded. We are left with c_{282} and c_{290}.

8.2. **Face equations and translation quotients.** For either homogeneous cubic c in (8.3)–(8.4), write

$$h(a) = c(a) + q(a) + \ell(a), \qquad q(a) = \sum_{1 \le i < j \le 9} q_{ij} a_i a_j. \tag{8.5}$$

An affine term $\ell(a) = b \cdot a + d$ can be removed without restricting the original problem: replacing h by $h + b \cdot a + d$ corresponds to replacing $f(x)$ by $f(x + b) + d$ and preserves the residue and forced-sign equations. We therefore take $\ell = 0$.

For a cubic h, formula (2.7) and the divisibility of seven-variable cubic weights imply

$$W_h(a) \equiv W_h(0) + 4 \sum_{i=1}^{9} M_{\{i\}}(h) a_i \pmod{32}.$$

In spectrum (7.1), each coefficient is 16 modulo 32. Thus the necessary conditions are

$$\mathrm{wt}(h) \equiv 8 \pmod{16}, \qquad M_{\{i\}}(h) \equiv 0 \pmod{8} \quad (1 \le i \le 9). \tag{8.6}$$

The nine face conditions are affine-linear in q. Indeed, use (3.1) modulo eight on an eight-variable face. A term that depends on one variable quadratic monomial survives only if it is accompanied by two disjoint cubic monomials, whose supports together partition the eight free variables into sizes $2 + 3 + 3$. No term involving two variable quadratic monomials can survive. Thus

$$Cq = b \quad \text{over } \mathbb{F}_2, \tag{8.7}$$

where the 9×36 matrix C and vector b are explicitly

$$C_{i,P} = \sum_{\substack{\{J,K\} \text{ unordered} \\ J \cup K \cup P = [9] \setminus \{i\} \\ |J| = |K| = 3}} c_J c_K \quad (P \subseteq [9],\ |P| = 2,\ i \notin P), \tag{8.8}$$

$$C_{i,P} = 0 \quad (i \in P), \qquad b_i = \frac{M_{\{i\}}(c)}{4} \pmod 2.$$

The divisions are integral because a degree-three function on eight variables has weight divisible by four.

Translating the input of h by $t \in V$ changes its quadratic part by the linear map

$$T(t)_{ij} = \sum_{k \notin \{i,j\}} c_{\{i,j,k\}} t_k. \tag{8.9}$$

The lower-degree terms created by the translation may again be removed. This is a genuine symmetry: $h(a+t)$ corresponds to $f(x) + t \cdot x$, so every relevant Walsh magnitude and the forced-sign obstruction are preserved. In particular $\operatorname{im} T \subseteq \ker C$.

Lemma 8.2 (Complete 18-dimensional transversals)**.** *For each of c_{282}, c_{290} the exact binary systems satisfy*

$$\operatorname{rank} C = \operatorname{rank} T = 9, \qquad CT = 0.$$

There is an explicit solution q_0 to (8.7) and an 18-dimensional subspace $U \leq \ker C$ such that

$$\ker C = \operatorname{im} T \oplus U.$$

Consequently every quadratic supplement satisfying the face equations is equivalent under phase translation to exactly one of 2^{18} representatives $q_0 + u$, $u \in U$. The remaining total-weight condition in (8.6) may be imposed on those representatives without losing any spectrum-(A) candidate.

Proof. The displayed ranks, kernel inclusions, particular solutions, and complements are verified by direct binary elimination using the integer-encoded data in Appendix A. The rank values imply $\dim \ker C = 36 - 9 = 27$, and the nine translation directions are independent; the given 18 complement vectors therefore give $\ker C = \operatorname{im} T \oplus U$. For any $q \in q_0 + \ker C$, write uniquely $q = q_0 + T(t) + u$. Translation by t changes the phase to one with quadratic supplement $q_0 + u$, up to removable affine terms. It preserves the condition $W_h \equiv 16 \pmod{32}$ and hence the remaining congruence $\operatorname{wt}(h) \equiv 8 \pmod{16}$. Every candidate is therefore represented. □

8.3. **The final finite certificate.** Enumerate the 2^{18} representatives for each cubic form. For every representative $h = c + q$, impose (8.6) and compute the integer Walsh transform $Q = W_h$. Only spectra with $|Q(x)| \in \{16, 48\}$ at every x need be considered. For each, the unique Boolean sign function forced by (7.2) is computed and its Walsh maximum is tested. All computations

are exact: the nine-stage fast Walsh transform uses only integer additions and subtractions, and the linear systems are over $\mathbb{F}_2$.

Finite certificate 8.3 (Exhaustive quotient enumeration)**.** The exact enumeration of the complete transversals gives

cubic top	states	$\mathrm{wt}(h) \equiv 8$	spectrum A	$\|W_f\|_\infty = 40$	$\|W_f\|_\infty = 48$
c_{282}	262144	128640	36960	18816	18144
c_{290}	262144	126592	43200	34560	8640
Total	524288	255232	80160	53376	26784

In particular, every spectrum-*(A)* phase surviving the necessary congruences has forced sign function satisfying $\|W_f\|_\infty \geq 40$. A separately executable verifier, `verify_rm19_two_cubics.py`, reconstructs C and T, checks all ranks and inclusions, visits every state, and validates a rejecting Walsh coefficient by an independent direct character sum.

Proof of Theorem 1.1. The lower bound $\rho(\mathrm{RM}(1,9)) \geq 242$ follows from the explicit constructions of Kavut and Yücel [6]. Suppose $\rho(\mathrm{RM}(1,9)) > 242$. The global upper bound 244 leaves a Boolean function of nonlinearity 243 or 244. By Theorem 1.2, existence of the first would imply existence of the second. We may therefore assume $\mathrm{NL}(f) = 244$.

Proposition 5.1 and Lemma 5.2 attach a cubic phase h with $\|W_h\|_1 = 10240$. Theorem 3.3 leaves spectra *(A)* and *(B)*. Proposition 6.1 excludes *(B)*. In *(A)*, the sign of f is forced by (7.2), and Proposition 7.1 together with Lemma 7.2 imposes $\omega^{\#\#} = \omega$ and $\Pi(\omega) = 0$. The published orbit classification and Certificate 8.1 leave only the representatives c_{282} and c_{290}. Lemma 8.2 covers all quadratic and affine supplements. Certificate 8.3 shows that the forced function always has $\|W_f\|_\infty \geq 40$, contrary to the necessary equality $\|W_f\|_\infty = 24$. This proves the upper bound 242 and the theorem. □

9. Conclusion

We have determined the covering radius of $\mathrm{RM}(1,9)$ by excluding the two possible nonlinearities above 242. The case of nonlinearity 243 is reduced to the endpoint 244 by the one-point rigidity theorem. More precisely, if $\mathrm{NL}(f) = 243$, then changing the value of f at the support sum $p(f)$ produces a function of nonlinearity 244. The proof uses integer interpolation of Walsh coefficients, rigidity of the resulting quadratic residue, and a sharp Walsh ℓ^1 bound for cubic Boolean functions. Consequently, the nonexistence of a function of nonlinearity 244 also excludes nonlinearity 243.

To rule out 244, we first combine Parseval's identity with congruences for integral Walsh spectra. These conditions force four possible signed Walsh values and yield a cubic residue phase attaining the extremal ℓ^1 bound. Of its two possible equality spectra, one is eliminated by comparing Walsh energies on complementary hyperplane slices. In the other, the phase determines the original function pointwise through its Walsh signs. A comparison of their

leading algebraic terms then gives the double-complement-pairing condition $\omega^{\#\#} = \omega$ together with $\Pi(\omega) = 0$. Using the published classification of alternating trilinear forms, we reduce the remaining possibilities to two explicit cubic top parts. For each one, the coordinate-face equations and input translations reduce the quadratic supplements to 2^{18} representatives. The exhaustive integer calculation excludes all $2 \cdot 2^{18}$ remaining cases. Hence neither 243 nor 244 is attainable; combined with the known construction of nonlinearity 242 [6], this proves

$$\boxed{\rho(\mathrm{RM}(1,9)) = 242.}$$

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Appendix A. Encoded data and reproducibility

We specify the finite inputs independently of any private project files. The 36 quadratic pairs are ordered lexicographically:

$$\begin{gathered} 12,13,14,15,16,17,18,19; \quad 23,24,25,26,27,28,29; \\ 34,35,36,37,38,39; \quad 45,46,47,48,49; \\ 56,57,58,59; \quad 67,68,69; \quad 78,79; \quad 89. \end{gathered}$$

Bit $j \in \{0,\dots,35\}$ represents the jth pair in this order. An integer therefore encodes any 36-bit quadratic coefficient vector. Dot products and ranks in this appendix are always computed over $\mathbb{F}_2$.

The cubic representatives are encoded by their monomial bit masks (the coordinate a_i is bit $i-1$):

$$\begin{aligned} c_{282} &: 19,25,28,44,69,138,268,304,448; \\ c_{290} &: 19,22,26,28,37,67,76,131,134,137,259,268,304,448. \end{aligned}$$

The particular quadratic solutions are

$$q_0^{(282)} = 16777216, \qquad q_0^{(290)} = 17725652992.$$

The following are explicit bases for the complementary subspaces U, listed in the same 36-bit encoding.

j	$u_j^{(282)}$	$u_j^{(290)}$
0	36507222016	2147483648
1	18253611008	5368709120
2	268435456	268435456
3	134217728	671088640
4	33554432	33554432
5	8388608	12582912
6	2097152	1048576
7	1048576	2883584
8	524288	25821184
9	65536	65536
10	16384	52076511232
11	4096	16384
12	4458496	25174016
13	536871168	16781312
14	128	25167872
15	64	4294967808
16	16777232	128
17	4	524320

The accompanying verifier computes every row of C from the coordinate-face weights rather than importing a stored matrix. It computes T from the cubic masks and verifies

$$\operatorname{rank} C = \operatorname{rank} T = 9, \quad Cq_0 = b, \quad CT = 0, \quad Cu_j = 0,$$
$$\operatorname{rank}\langle T_1, \ldots, T_9, u_0, \ldots, u_{17}\rangle = 27.$$

A candidate representative has quadratic word $q = q_0 + \sum_j \epsilon_j u_j$, where $\epsilon \in \mathbb{F}_2^{18}$. The Walsh algorithm uses the exact butterfly $(r, s) \mapsto (r + s, r - s)$ in all nine coordinates. For all 80160 equality-spectrum states, it also evaluates one selected rejecting character sum directly from the 512 truth bits, independently of the transform that selected the frequency. On a successful run, the verifier prints

```
class 282: states=262144, odd_parity=128640, spectrum_A=36960,
           forced_peaks={48: 18144, 40: 18816}
class 290: states=262144, odd_parity=126592, spectrum_A=43200,
           forced_peaks={48: 8640, 40: 34560}
PASS: both quotient enumerations, rank tests and direct rejecting sums.
```

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